

Research Direction, Novelty & Evidence

What this thesis actually proves, and where it stands against the current literature (verified via web search, July 2026)

Generated 2026-07-09 · Synthesizes Phases 1-5 empirical results against a fresh literature check

Refined contribution statement. This thesis builds and empirically validates **ICS-NAD-Gym**, the first sequential-decision RL environment grounded in real, brand-diverse (ABB/Schneider/Siemens), flow-level ICS network telemetry, together with **an LLM-guided, policy-invariant reward-shaping mechanism** that measurably improves the resulting policy's generalization to held-out capture sessions — demonstrated with a leak-free split, a 10-classifier supervised baseline, and a 3-seed matched-budget statistical comparison. As of this writing, no published work combines these specific elements (Section 2), and the two problems this thesis addresses empirically — F1 vs. operational-cost mismatch, and train/test leakage in flow-based NIDS datasets — are independently recognized as open, active concerns in the current literature (Section 3), which strengthens rather than weakens the contribution: this isn't a strawman problem invented for the thesis.

1. What Was Actually Built and Proven (not just proposed)

Claim	Evidence
A real, working RL environment can be built directly from published ICS flow telemetry, with a stateful, cost-asymmetric response action space	ICSNADGym : verified via <code>check_env</code> + 4-policy sanity floor (oracle strictly dominates); Phase 2
The dataset's original random-split evaluation methodology was unsound, and a proper leak-free split reveals a real, previously-hidden generalization gap between classifier families	KNN/SVC/ANN/ELM: val F1≈0.9996 → test F1≈0.831 (massive collapse); tree ensembles (GBDT etc.) generalize cleanly, GBDT even improves val→test; Phase 4
A classifier's F1 score does not reliably predict its value as an actual sequential response policy under a realistic asymmetric cost structure	SVC and GaussianNB (F1≈0.77, "workable" by ML standards) scored <i>negative mean reward</i> as in-simulator policies; Phase 5 addendum
LLM-guided, policy-invariant reward shaping (archetype clustering + local-LLM risk scoring, $\Phi(s)$ potential function) can be built with zero LLM calls in the RL training hot loop, and measurably improves policy generalization	3-seed, matched-2M-step-budget, identical-hyperparameter comparison: mean test F1 0.880→0.920, 2 of 3 seeds show a clear, similar-magnitude gain (+0.058 to +0.064), variance across seeds drops under shaping (0.043→0.012); Phase 5 + Phase 5c

2. Novelty Check Against Current Literature (verified July 2026)

2.1 The dataset

No published work uses the ICS-NAD dataset (Scientific Data/Nature, 2026) in combination with any reinforcement learning environment — a direct search for this pairing returns nothing. This RL environment is the first built from it.

2.2 The closest adjacent work: L2M-AID

[L2M-AID](#) (2025, also published at [IEEE](#)) remains the nearest published system combining LLMs with RL for industrial cyber-physical defense. Direct comparison:

	L2M-AID	This thesis
Where the LLM acts	Translates raw telemetry into a semantic state representation for the RL agent to observe	Scores flow archetypes offline to build a potential function that shapes the reward , never touching the observation
RL formulation	Multi-agent MAPPO, team of LLM-driven agents	Single-agent RecurrentPPO
Data	SWaT (water-treatment physical-process sensors) + synthetic MITRE ATT&CK-for-ICS attack data	Real captured ICS network flow telemetry (ABB/Schneider/Siemens PLCs), no synthetic attacks
Reward-invariance guarantee	Not the focus — reward balances security vs. process stability heuristically	Potential-based shaping is <i>provably</i> policy-invariant (Ng/Harada/Russell 1999), giving a rigorous way to isolate the LLM's specific contribution via ablation

A second closely related line — [LLM-based reward design for autonomous cyber defense](#) — exists only for enterprise-IT benchmarks (CAGE-style challenges), not ICS/OT, and does not use the archetype-clustering + potential-function mechanism used here. [Heuristic extraction from LLMs for reward shaping](#) is architecturally closest in spirit (LLM produces a shaping signal) but is domain-general and was never applied to network security or ICS telemetry.

Conclusion: the specific combination — potential-based, LLM-archetype-scored reward shaping, applied to a real, flow-level ICS-NAD-Gym RL environment, with an ablation-clean multi-seed statistical validation — has not been published elsewhere as of this search.

3. Supporting Evidence That the Problems Addressed Are Real (not invented)

3.1 F1/accuracy vs. operational cost mismatch

Independently confirmed in the literature: *"machine learning-based NIDSs typically optimize uniform metrics such as accuracy and F1-score, overlooking the asymmetric cost structure of real-world security operations, where a missed attack (FN) far outweighs a false alarm (FP)."* A dedicated [cost-sensitive threshold-optimization paper](#) (2026) addresses exactly this gap for XGBoost using a 10:1 FN:FP cost ratio. This validates Phase 5's empirical finding (SVC/GaussianNB: workable F1, negative reward) as an instance of a recognized, active problem — and positions this thesis's solution (full sequential RL framing with a stateful, asymmetric reward) as a more general answer than post-hoc threshold tuning on a static classifier.

3.2 Train/test leakage in flow-based NIDS evaluation

Independently confirmed: *"a naive random splitting approach is not recommended, as [flow] samples often overlap in time and/or space"* with test data, and the field's own recommended fixes are exactly **group-by-flow-tuple** and **time-based** splitting — [surveyed directly in a 2026 paper on NIDS dataset limitations](#). Phase 1's file-level split (the ICS-NAD-specific analogue of "group-by-session" splitting) is a direct, dataset-specific instance of this recommended practice — applied here for the first time to ICS-NAD, whose original publication used a naive random split.

4. What This Means for the Thesis's Framing

The strongest, most defensible framing is not "we invented a new problem" — it's **"we built the first RL testbed on this dataset, used it to empirically demonstrate two problems the field already recognizes as important (F1/cost mismatch, split leakage) are real and severe on this specific dataset, and validated a novel LLM-guided shaping mechanism — not previously applied in this way — that measurably helps."** Citing the F1-mismatch and split-leakage literature (Section 3) makes the motivation stronger, not weaker: it shows the thesis is solving a problem the field already cares about, on a concrete, non-toy testbed, with a mechanism nobody has yet tried in this configuration (Section 2).

4.1 Recommended thesis contribution statement (final form)

"We present ICS-NAD-Gym, the first Gymnasium-compatible RL environment built from real, brand-diverse ICS network flow telemetry, and LLM-ShAId, a potential-based reward-shaping mechanism using LLM-scored flow archetypes. Using a leak-free, file-level train/test split — itself revealing that several standard ML classifier families catastrophically overfit under the dataset's original (leaky) evaluation methodology — we show that a plain RL policy already

outperforms most static classifiers as a sequential response policy under a realistic asymmetric cost structure, and that LLM-guided reward shaping provides a consistent, multi-seed-validated improvement in the policy's generalization to unseen capture sessions, at no cost to training-distribution performance."

4.2 Immediate next steps that would most strengthen this claim

Priority	Action	Why
High	2 more seeds (5 total) at matched 2M-step budget	n=5 gives the Wilcoxon test a chance to reach conventional significance if the true effect is as consistent as seeds 0/1 suggest
Medium	Write up the L2M-AID delta explicitly as related work in the thesis text	Pre-empts the most likely reviewer/examiner question ("isn't this just L2M-AID?") with a direct, sourced comparison
Medium	Cite the cost-sensitive-threshold and split-leakage papers found here directly in the thesis's motivation section	Converts "empirical finding" into "empirical confirmation of a literature-recognized problem," which is a stronger claim to a reader
Lower	Detection-latency metric (already designed, not yet measured)	A capability metric no static classifier can report at all — a clean, additional differentiator beyond F1/reward

5. Sources

- [L2M-AID: Autonomous Cyber-Physical Defense by Fusing Semantic Reasoning of LLMs with Multi-Agent RL](#) (arXiv 2510.07363 / IEEE)
- [Large Language Model-Based Reward Design for DRL-Driven Autonomous Cyber Defense](#) (arXiv 2511.16483)
- [Extracting Heuristics from Large Language Models for Reward Shaping in RL](#) (arXiv 2405.15194)
- [LLM-Guided Incentive Aware Reward Design for Cooperative Multi-Agent RL](#) (arXiv 2603.24324, under review)
- [Cost-Sensitive Threshold Optimization for Network Intrusion Detection: A Per-Class Approach with XGBoost](#) (Electronics, 2026)
- [Network Intrusion Datasets: A Survey, Limitations, and Recommendations](#) (arXiv 2502.06688)
- [Policy-Invariant Reward Shaping](#) (background/theory reference)